

*** Temporal Attentive Graph Networks (TAGN) for Financial Surveillance (Zhang et al., 2026)***

Dataset:

Micro-scale (market microstructure): OHLCV price/volume data, Amihud Illiquidity Ratio, and Realized Volatility computed over 5-minute high-frequency windows

Sentiment-scale: Daily news headlines encoded via FinBERT into sentiment scores

Macro-scale: VIX index and Treasury yield curve slope

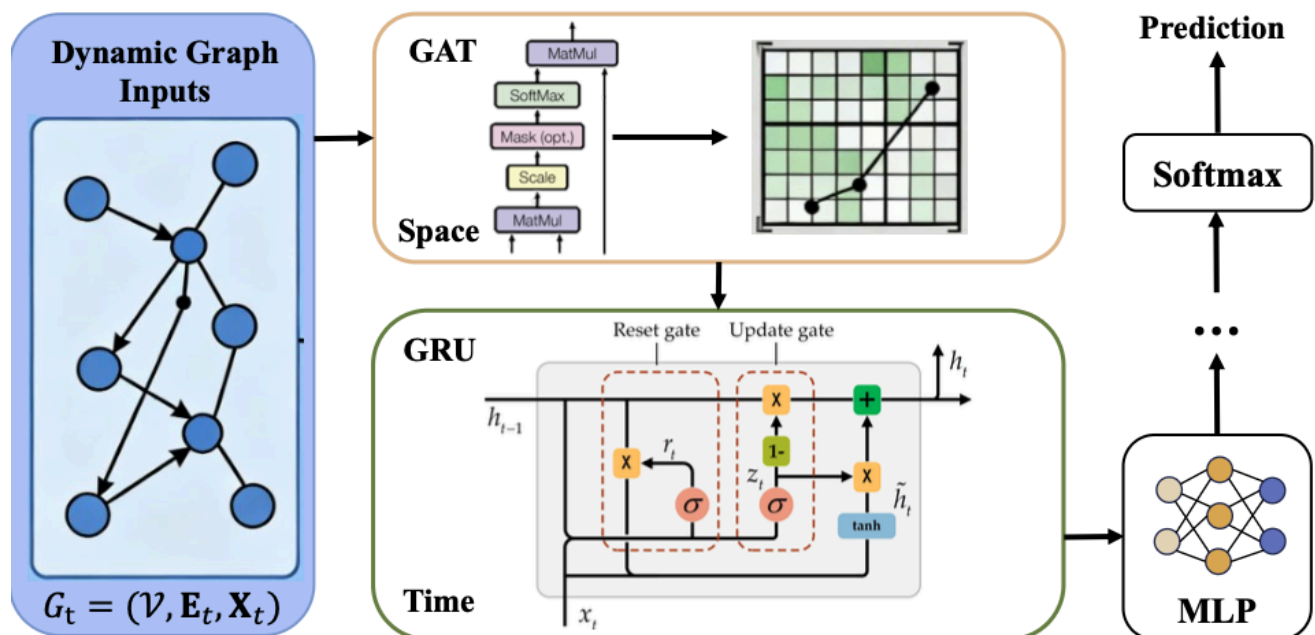
Relational data: Supply chain linkages (from FactSet), institutional co-holdings (from SEC Form 13F filings), and price correlation graphs

OHLCV data is pretty available to get.

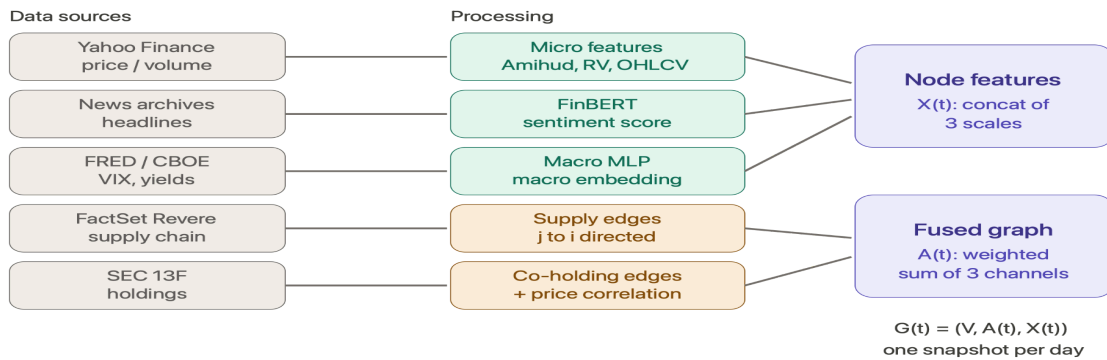
FinBERT is free to use and new data is access from different sources.

Only expensive data is supply chain linkages or factset. (**FactSet ranges roughly \$4,000–\$50,000 per user/year depending on modules; Workstation tiers run Basic ~\$333/user/year, Standard ~\$1,000/user/year, Premium ~\$2,500/user/year, with median actual contracts around \$25,000/year**)

per-stock, per-day crash forecast
per-stock, per-day crash forecast
execution ;

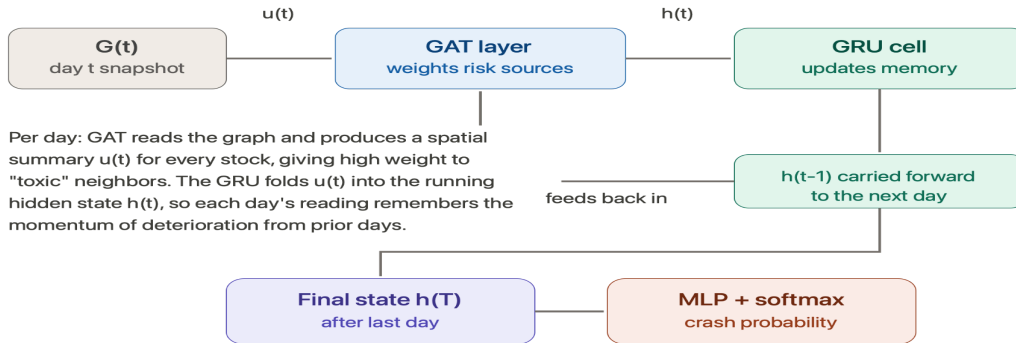


Stage 1 — raw data to graph snapshot



Each trading day becomes one graph snapshot: nodes are S&P 500 stocks, node features hold the three feature scales, and edges hold the three fused risk-transmission channels. The full history is an ordered sequence of these snapshots $G(1) \dots G(T)$, which feeds Stage 2.

Stage 2 — encoder loops over the snapshot sequence

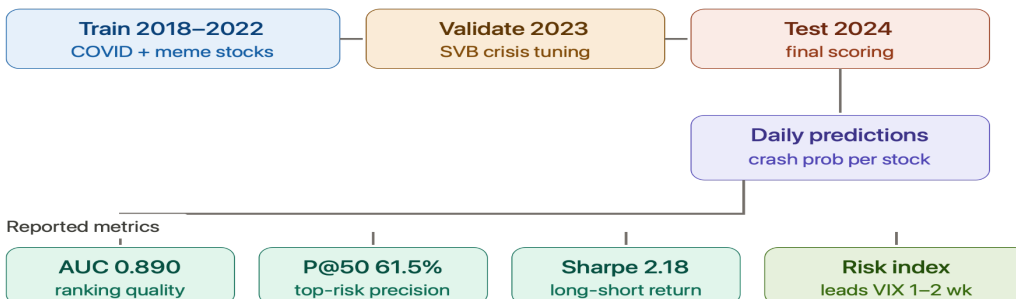


The loop repeats for every day in the input window. The hidden state from the final day, $h(T)$, is the model's full spatiotemporal read on each stock. An MLP plus softmax turns that into a probability that the stock will crash (drop more than 10%) within the next 5 trading days.

Training uses Focal Loss so rare crash days dominate the gradient instead of being drowned out.

Stage 3 — train, test, and produce results

Chronological split (no look-ahead)



The model only ever sees the future through the test year. Daily crash probabilities are scored four ways: AUC and Precision@50 measure prediction quality, the Sharpe ratio measures a long-short trading strategy built on those scores, and aggregating all 500 probabilities into one number gives the TAGN-Risk Index — which the paper shows rising before the VIX during stress.

<https://github.com/dcpengjin/temporal-attentive-graph-networks-for-financial-surveillance/tree/main>

This experiment excluded some important factor for real market. When we recreate this we have to keep in mind:

- 1) We need to include transection cost, slippage etc.
- 2) We need to apply walk-forward to see the performance and in the paper they didn't included it.

Stage 1 — Reproduce the artifact (1–2 weeks)

Goal: confirm the published thing actually works as published, on their data.

- Pull their Gitee repo, run it on their Zenodo data, hit their in-sample numbers.
- Don't build anything of your own yet.

Kill-gate: Can you reproduce $AUC \approx 0.89$ in-sample? If no → the artifact is broken or underspecified, and nothing downstream is trustworthy. Stop and reassess before investing more.

Why first: it's the cheapest possible test, and debugging your own reimplementaion against a target you can't even reproduce is hopeless.

Stage 2 — Rebuild clean, on data you control (2–4 weeks)

Goal: a leakage-free reimplementaion on independent, point-in-time data.

This is where the real discipline lives. The order *within* this stage matters:

1. **Get point-in-time S&P 500 membership first.** Before any modeling. If your universe is "today's constituents run backward," everything after is contaminated. This is the most common silent killer.
2. Build the feature pipeline ($X(t)$) and the label generator separately, then **audit for leakage** — every feature must use only information known on or before the prediction date; normalization must never use full-dataset statistics.

3. Build the graph construction. If you don't have FactSet, build with correlation + 13F graphs only and *note the missing channel* — don't fake it.
4. Wire in the GAT+GRU encoder and Focal Loss.

Kill-gate: Does your clean reimplementation produce *sane* in-sample numbers (not necessarily 0.89, but clearly better than chance)? And does a deliberate leakage-audit come back clean?

Stage 3 — Walk-forward validation (1–2 weeks, the decisive stage)

Goal: find out if the signal is real or a single-regime fluke.

- Rolling windows: train on years 1–3, test year 4; roll forward; retrain each fold; evaluate only on untouched future data.
- Log AUC, F1, P@50 **per fold**, not just averaged.

Kill-gate (decide the threshold now): Does it beat baselines *consistently across folds* — not on average, but in most/all windows? Critically, run **plain GRU and XGBoost on the same features** as baselines. If TAGN doesn't clearly beat them, the expensive graph machinery isn't earning its keep, and you should either simplify or stop.

This is the stage that kills most strategies. That's the point. It's cheap and decisive — do it before you spend a dollar on FactSet.

Stage 4 — Realistic-cost backtest (1–2 weeks)

Goal: find out if the *trading* result survives the real world. (Skip this entirely if your goal is monitoring, not trading.)

- Add transaction costs, slippage.
- Apply the **borrow-availability filter** — drop hard-to-borrow names from the short book. The riskiest decile (what the model shorts) is exactly what's often unborrowable.
- Test a turnover-capped, slower-rebalancing version.

Kill-gate: Is the cost-adjusted, borrow-filtered Sharpe still positive and meaningful across folds? A realistic survivor lands maybe 0.5–1.0. If it still shows 2.18, you have a bug or leakage — investigate, don't celebrate.

Stage 5 — Forward paper-trade (3–6+ months, only if Stages 1–4 pass)

Goal: the only uncontaminated test — run it live with no money.

- Catches what backtests can't: late filings, revised data, the 13F lag biting in a fast move, pipeline breakage.

Kill-gate: Does live-forward performance track the backtest, even roughly? Divergence here means hidden look-ahead somewhere upstream.

Cost:

S&P 500 daily price/volume Yahoo Finance (yfinance/bloomberg) **Free**

5-min intraday (for Realized Volatility) Polygon.io paid plans start at \$199/month for serious usage; SEC cheaper tiers from ~\$29/mo

News headlines GDELT (free) or NewsAPI **Free**, or ~\$449/mo for NewsAPI business tier

FinBERT model Hugging Face **Free**

VIX + Treasury yields FRED / CBOE **Free**

Institutional holdings SEC EDGAR 13F **Free**

Supply chain links FactSet / WRDS **\$0 via university**, otherwise \$25k+/yr

Point-in-time index membership CRSP / paid The hidden cost — often ~\$0 via university, else Bloomberg

Computation can be done in google colab for 50\$ with the pro or with google VM which will be a little costly but efficient for running with a larger memory and GPU access.